

Given Γ in G with parameterization $\mathbf{q}(t)$, $t \in [\alpha, \beta]$, we consider the curve Γ^ε with parameterization $\mathbf{q}^\varepsilon$ defined by

$$\mathbf{q}^\varepsilon(t) = \mathbf{q}(t) + \varepsilon \boldsymbol{\eta}(t), \quad t \in [\alpha, \beta], \quad (2.3)$$

where $\boldsymbol{\eta}(t)$ satisfies $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = \mathbf{0}$. Thus $\boldsymbol{\eta}$ is a C^∞ parameterized curve linking $\mathbf{0}$ to $\mathbf{0}$. In defining variations in this way, we are using, implicitly, the fact that we can add together functions in C^∞ and multiply them by scalars (e.g. ε) and remain in C^∞ . We also make use of the intuitive concept that (2.3) defines $\mathbf{q}^\varepsilon$ in such a way that it is “close” to $\mathbf{q}$ when ε is small. This can be made precise by a little more elaboration, but we forego this here. Effectively, we are using our understanding of C^∞ as a *normed function space* to restrict attention to variations of the base curve Γ in a particular “direction.”

Using a Taylor series expansion of the Lagrangian,³ we have

$$\begin{aligned} \mathcal{A}_L(\Gamma^\varepsilon) - \mathcal{A}_L(\Gamma) &= \int_{\alpha}^{\beta} [L(\mathbf{q}(t) + \varepsilon \boldsymbol{\eta}(t), \dot{\mathbf{q}}(t) + \varepsilon \dot{\boldsymbol{\eta}}(t)) - L(\mathbf{q}(t), \dot{\mathbf{q}}(t))] dt, \\ &= \int_{\alpha}^{\beta} \left[\varepsilon \left(\frac{\partial L}{\partial \mathbf{q}}(\mathbf{q}(t), \dot{\mathbf{q}}(t)) \boldsymbol{\eta}(t) + \frac{\partial L}{\partial \dot{\mathbf{q}}}(\mathbf{q}(t), \dot{\mathbf{q}}(t)) \dot{\boldsymbol{\eta}}(t) \right) + \mathcal{O}(\varepsilon^2) \right] dt. \end{aligned}$$

Hamilton’s principle states that the natural motion of the system described by the Lagrangian L is a stationary point of the classical action which implies that the $\mathcal{O}(\varepsilon)$ term above should vanish for any smooth variation $\boldsymbol{\eta}(t)$ with $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = \mathbf{0}$, i.e.,

³In the multidimensional setting, Taylor’s theorem states that given a C^{k+1} function $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and a point $\mathbf{z}_0$, we have

$$\begin{aligned} f(\mathbf{z}) - f(\mathbf{z}_0) &= \nabla f(\mathbf{z}_0) \cdot (\mathbf{z} - \mathbf{z}_0) + f^{(2)}\langle \mathbf{z} - \mathbf{z}_0, \mathbf{z} - \mathbf{z}_0 \rangle + f^{(3)}\langle \mathbf{z} - \mathbf{z}_0, \mathbf{z} - \mathbf{z}_0, \mathbf{z} - \mathbf{z}_0 \rangle \\ &\quad + \dots f^{(k)}\langle \mathbf{z} - \mathbf{z}_0, \mathbf{z} - \mathbf{z}_0, \dots, \mathbf{z} - \mathbf{z}_0 \rangle + \mathcal{O}(\|\mathbf{z} - \mathbf{z}_0\|^{k+1}) \end{aligned}$$

where $\nabla f = \partial f / \partial \mathbf{z}$ is the gradient, i.e. a vector with m components, $f^{(2)}$ is the $m \times m$ Hessian matrix of f (the matrix whose ij component is $\partial^2 f / \partial z_i \partial z_j$), and $f^{(2)}\langle \mathbf{u}, \mathbf{v} \rangle$, $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, represents the quadratic form $\mathbf{u}^T f^{(2)} \mathbf{v}$. In a similar way we interpret $f^{(3)}$ as a tensor which we can think of as a $m \times m \times m$ triply-indexed array, the ijk element being $\partial^3 f / \partial z_i \partial z_j \partial z_k$ and

$$f^{(3)}\langle \mathbf{u}, \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m (\partial^3 f / \partial z_i \partial z_j \partial z_k) u_i v_j w_k.$$